Refinements to Rceattle for eastern Bering Sea pollock

Rceattle 5.8.1 working paper for the 2024 assessment configuration

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Abstract

The custom AD Model Builder (ADMB) stock assessment model developed for eastern Bering Sea walleye pollock has a long history of application. For a variety of reasons, including general recommendations from the 2025 Center for Independent Experts (CIE) reviews, the custom ADMB code has been successfully ported to R Template Model Builder (RTMB). That conversion provides access to advanced RTMB features and makes the underlying code easier to understand and modify. Similarly, adopting more generic platforms, such as Rceattle, can provide access to more modern software developments. In this study, we tailored Rceattle code and configurations so that Rceattle can closely mimic the custom ADMB/RTMB assessment code. The Rceattle configuration uses the supported AFSC multinomial composition likelihood and a non-parametric selectivity form. We also evaluated a two-dimensional, first-order autoregressive (AR1) selectivity sensitivity; it did not converge and remains a research topic. An exploratory dynamic structural equation modeling (DSEM) analysis shows how recruitment covariates can be compared by cohort, separately from the final assessment configuration.

Rceattle WORKING PAPER

1 Executive Summary

Building from the ADMB-to-RTMB conversion, this analysis evaluates whether Rceattle can be configured to reproduce the main features of the 2024 eastern Bering Sea pollock "PM" assessment while providing a model that is easier to extend. This 5.8.1 update focuses on the Rceattle configuration, comparison with the modified ADMB model, model diagnostics, and projection inputs.

When the Rceattle parameters and fishery selectivity are set to the same values used in the ADMB model, Rceattle reproduces numbers-at-age, spawning biomass, and predicted catch to numerical precision. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. This result establishes agreement in the population dynamics when both implementations use the same parameter values.

When Rceattle was fit to the data, convergence was reasonable but estimability was somewhat unstable for the annual selectivity deviations. Relative to the ADMB fit, mean absolute differences were 1.3% for spawning biomass, 0.3% for recruitment, and 1.0% for total biomass. These small differences are concentrated in the earliest years; the remaining uncertainty is primarily estimability of the annual selectivity deviations.

The 2D age-by-year AR1 selectivity sensitivity was included as an initial diagnostic attempt. Results appeared broadly similar to the non-parametric constrained selectivity fits. However, because some parameters reached bounds, the model failed to converge. Closer examination of this selectivity form will continue as a research topic.

One-step-ahead residuals and nine terminal-year-consistent retrospective peels have been generated from the Rceattle configuration. Aggregate composition residual dispersion is lower than the standard-normal reference. Mohn's rho is 0.274 for spawning biomass, 0.137 for total biomass, 0.139 for recruitment, and -0.157 for fishing mortality. Forty-eight of 50 jitters returned to the best solution; two found a substantially poorer mode. These results focus subsequent work on selectivity identification, retrospective bias, and observation-model sensitivity.

2 Purpose and Scope

The September 2026 review focuses on whether the Rceattle configuration has sufficient documentation for Plan Team evaluation. This working paper serves that focused purpose rather than the broader purpose of a full Stock Assessment and Fishery Evaluation (SAFE) chapter.

This working paper therefore records:

- what was run from those development files;
- how the Rceattle outputs compare with the modified ADMB reference;
- which diagnostics are available now;
- which sensitivities remain research-only; and
- what work would help structure the next evaluation pass.

The report uses the Alaska Stock Assessment Report (`asar`) package for glossary conventions and accessibility guidance (Schiano et al. 2026). Those practices helped prepare the Quarto source for Section 508 review, including semantic headings and tables, descriptive figure captions, and alternative text. They support accessibility but do not replace final manual and agency-approved compliance checks.

Standard SAFE report sections such as background material, fishery history, ecosystem narrative, reference-point tables, and harvest advice are outside the scope of this document. Projection calculations are included only to test the export and use of the Rceattle results.

3 Rceattle Configuration

3.1 Model Inputs

The Rceattle configuration uses the prepared workbook `Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx`. The data builder retains every age from 1 through 15. Fishery compositions use nominal sample sizes; bottom trawl survey (BTS) and acoustic-trawl survey (ATS) sample sizes use the integer values aligned with the modified ADMB model. Age-1 survey selectivity remains separately parameterized from the age-2–15 curve, and each likelihood contribution evaluates the complete 15-age composition. Compositions use the supported `MultinomialAFSC` likelihood. We added a robustness constant of 0.001 to both observed and expected proportions before evaluating the likelihood. This prevents `log(0)` and limits the influence of observations in bins with expected proportions near zero.

The run uses Rceattle version 5.8.1 from upstream commit 7bb694788012, with the model configuration synchronized to `Rceattle-models` commit 3dc99ddfbf9e.

The next-assessment update script review is complete, and execution awaits complete new-year observations. The script duplicates terminal records and places placeholders for new catch, composition, and survey observations. The script serves as a template until complete new-year observations are available; after population, it creates a complete assessment workbook.

3.2 Selectivity

The configuration uses `NonParametricPM` fishery selectivity with annual deviations. When fit to the data, the model reached a small maximum gradient. The Hessian condition number remained high, with its least-identified direction concentrated in `sel_coff_dev`; thus, estimability of the annual selectivity deviations remains somewhat unstable.

The time-varying selectivity penalties use explicit standard deviations on log-selectivity random-walk increments. The fishery and its shared CPUE selectivity block use an SD of 0.500, ATS and AVO use 0.138, and BTS uses 1.000. Thus, the fishery curve has appreciable annual flexibility, whereas the ATS and AVO curves receive stronger temporal smoothing. The fishery specification also retains the ADMB-matched curve-shape penalty weights (12.5, 1/60, and 1) and an average-selectivity centering weight of 10. These values are penalty coefficients rather than SDs.

The fitting sequence, composition-likelihood sensitivity, and 2D age-by-year AR1 development work are documented in the appendix.

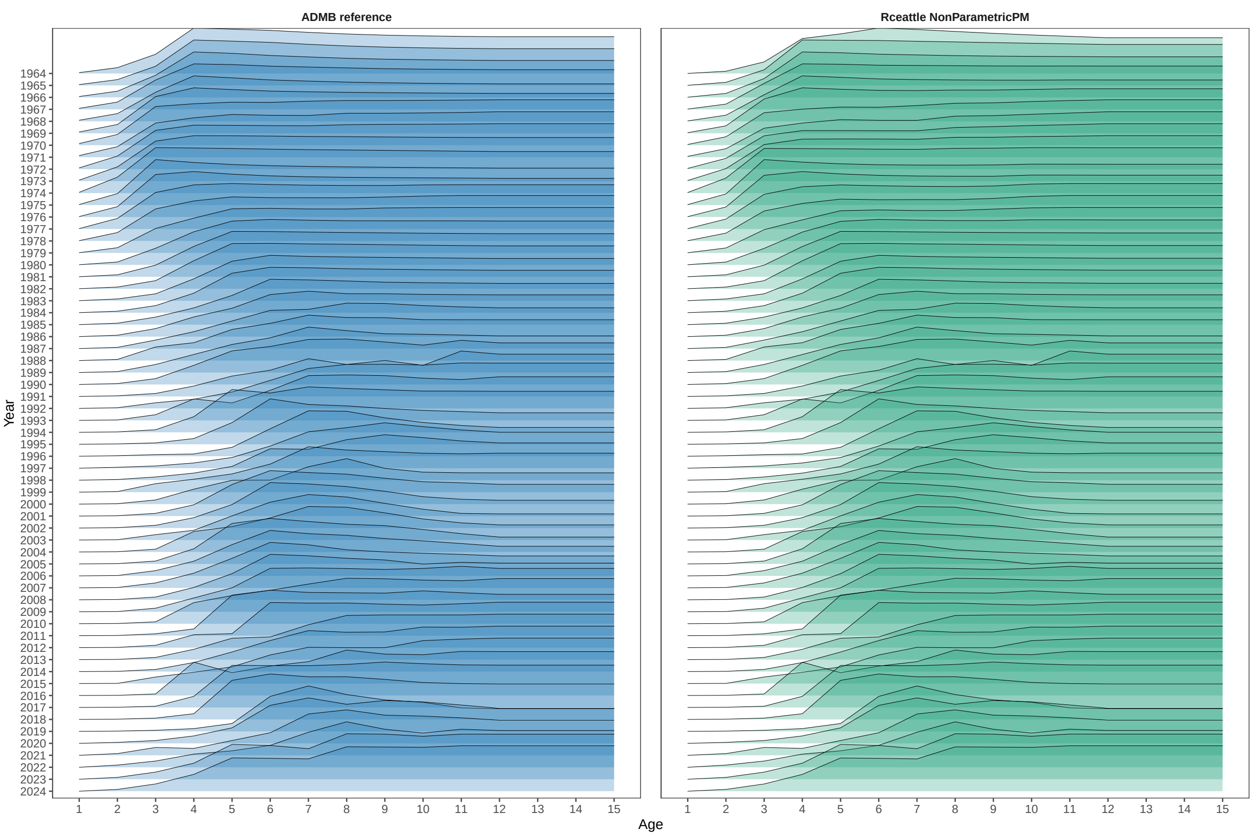

Figure 1: Relative fishery selectivity by year for the ADMB model and Rceattle configuration. Each annual curve is normalized to a maximum of one.

4 ADMB Comparison Results

When the Rceattle parameters and fishery selectivity are set to the ADMB values, the two programs give nearly identical population calculations. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. Annual Rceattle-to-ADMB ratios for both quantities remain extremely close to one (Figure 2).

The reported Rceattle likelihood contributions are 140.191 for indices, -123.761 for catch, 3572.874 for age compositions, and 98.379 for recruitment deviations. Their sum is 3704.588. The ADMB `tot_like` and `dat_like` values are 736.724 and 459.931, respectively, but cannot be compared directly because the two programs include different likelihood terms, penalties, and constants. The ADMB maximum absolute gradient is 3.47e-04; no Rceattle model-parameter gradient is available from this comparison because Rceattle does not estimate parameters in this run.

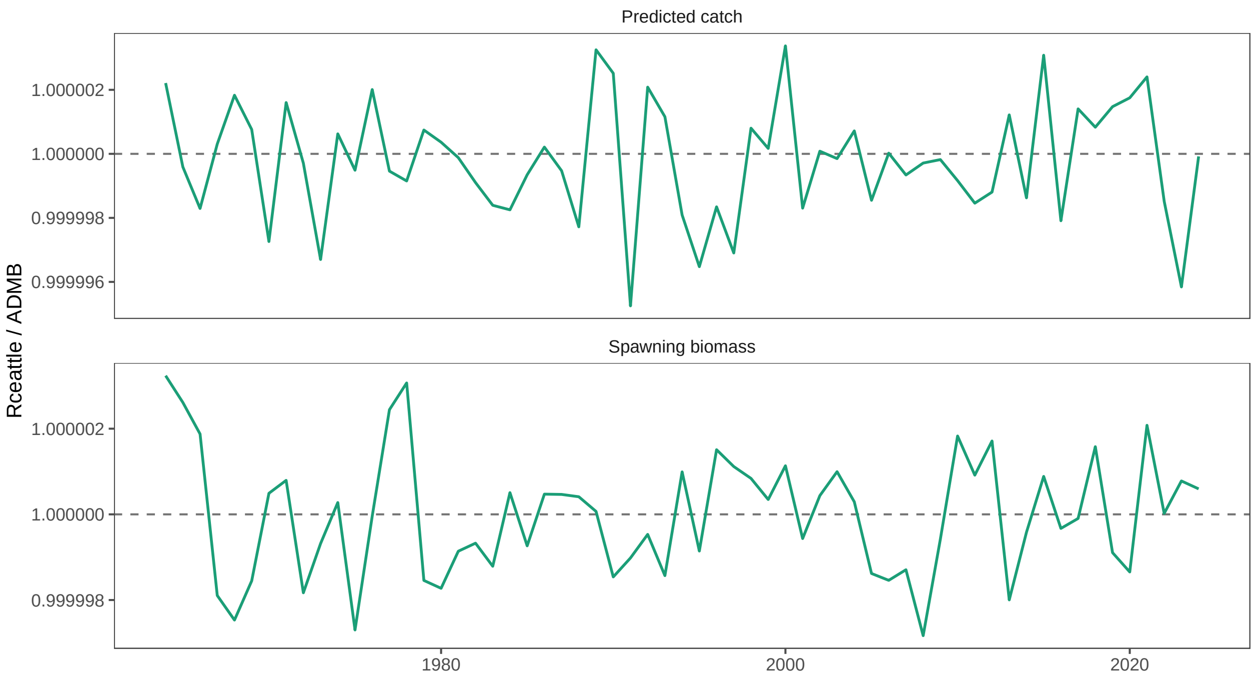

Figure 2: Rceattle-to-ADMB ratios for spawning biomass and predicted catch when Rceattle uses the ADMB parameter values.

Table 1: Agreement between fitted Rceattle trajectories and the modified ADMB reference. Correlation is the Pearson correlation across annual estimates. Mean absolute (%) and maximum absolute (%) are the average and largest absolute annual percentage differences, respectively. The 2024 difference (%) is the signed terminal-year difference of Rceattle relative to ADMB; negative values indicate that the Rceattle estimate is lower.

Quantity	Correlation	Mean absolute (%)	Maximum absolute (%)	2024 difference (%)
Rceattle configuration				
Recruitment	1.00	0.29	2.04	-0.85
Spawning biomass	1.00	1.29	9.95	-0.18
Total biomass	1.00	1.03	8.78	-0.18

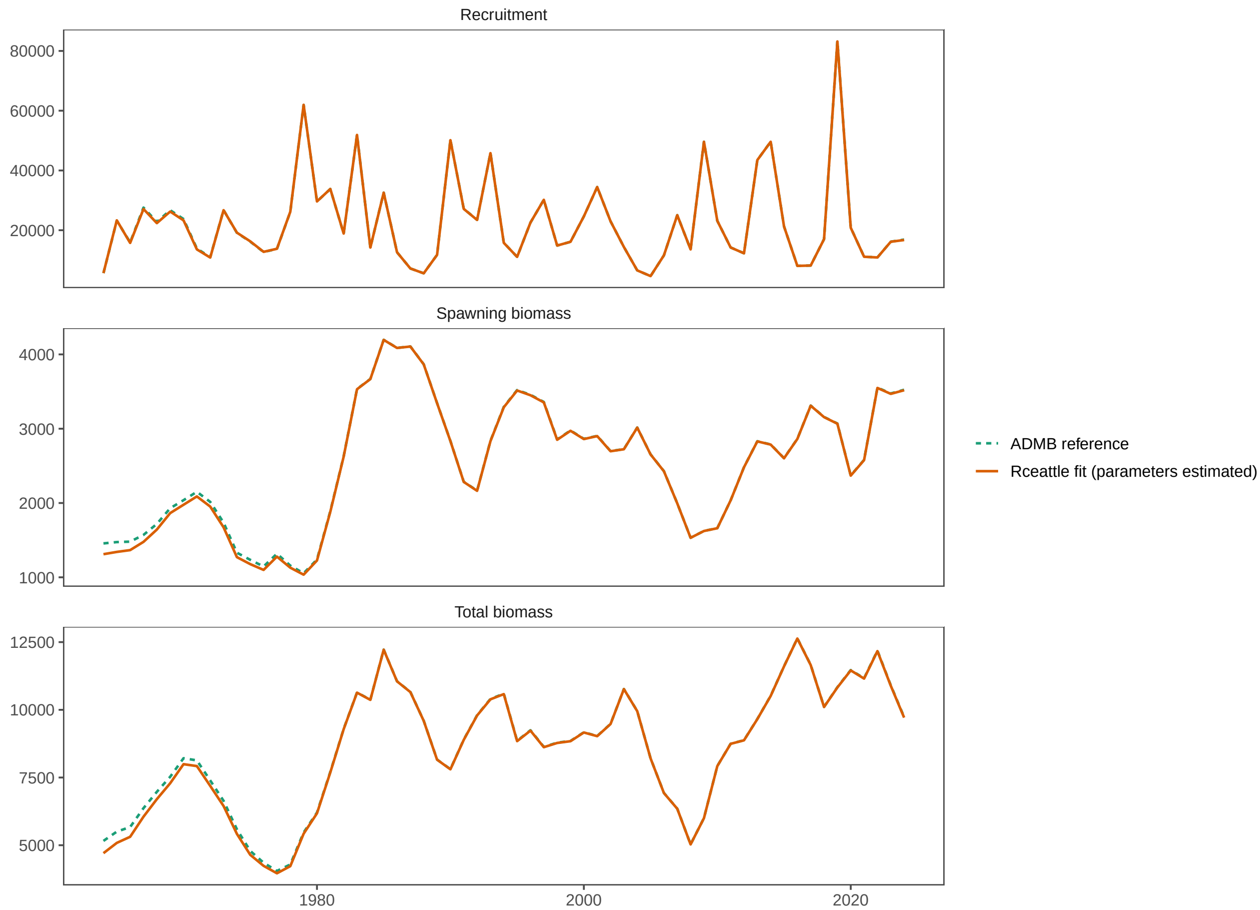

Figure 3: ADMB and fitted Rceattle trajectories for spawning biomass, recruitment, and total biomass. The Rceattle series is from the fit in which its parameters were estimated.

When Rceattle estimates its parameters, the fitted trajectories differ most from ADMB during the early period and in the terminal recruitment estimate (Figure 3). These trajectories are distinct from the run in which Rceattle parameters are set to the ADMB values. The fitted trajectories remain highly correlated with the ADMB reference, with relatively small mean annual differences (Table 1). Development sensitivities are documented in the appendix.

The Rceattle fit has a small maximum gradient and a positive definite Hessian, but the Hessian-conditioning warning is substantive. The remaining weak direction is almost entirely in the time-varying fishery selectivity coefficients. This issue should be resolved or clearly bounded before the fit is considered an operational assessment candidate.

5 Diagnostics

The numerical convergence checks for the fitted Rceattle model, including the objective, maximum gradient, and Hessian status, indicated convergence (Table 2). The fit and status of the other diagnostic products are provided in Table 3.

Table 2: Stored Rceattle convergence checks for the evaluated configurations.

Model	Status	Objective	Max gradient	Weakest component	PD Hessian
Rceattle fit	WARN	713.676499	0.000044	sel_coff_dev	TRUE

Table 3: Diagnostic status for the September 2026 review.

Diagnostic	Current status	Review use
ADMB-parameter comparison	Completed and saved in the ADMB-to-Rceattle comparison outputs.	Primary numerical agreement check.
Rceattle fit	Completed with WARN status from Hessian conditioning.	Primary Rceattle comparison fit to the data.
2D AR1 selectivity sensitivity	Received a FAIL convergence status, an unsuccessful Hessian inversion, and parameter-bound warnings.	Requires further research before it can be considered as an alternative selectivity form.
OSA residuals	Completed for 1,864 residuals and saved in RDS outputs.	Useful, but analytical and exactly reproduced index groups need interpretation.
Rceattle retrospectives	Nine peels completed for the Rceattle fit with OK status.	The revised peels preserve data availability and constrain terminal selectivity parameters to the preceding year.
Legacy ADMB retrospectives	Existing ADMB retro.rep files are present.	Provides historical context distinct from the Rceattle retrospective peels.
Rceattle 5.8.1 self-tests	Fifty initial-start and 50 estimated-start refits completed with OK status.	Supports the revised positive-index simulation workflow; it does not resolve observed-data multimodality.
Jitters and likelihood profile	Forty-eight of 50 jitters returned to the best mode; two WARN fits reached an inferior mode. The age-3+ M profile was shallow with its grid minimum at 0.35.	Documents multimodality and weak information about natural mortality.

5.1 Rceattle 5.8.1 self-test validation

Rceattle 5.8.1 changed simulation of natural-scale normal and multivariate normal survey indices. The fitted likelihood remains an untruncated normal; the simulator redraws non-positive values because `data_check()` excludes them. The resulting self-test therefore uses a normal distribution truncated at zero whenever rejection occurs.

The ADMB-to-Rceattle comparison was rerun first with Rceattle 5.8.1. Its saved summary gives mean

Table 4: Rceattle 5.8.1 self-test and paired-restart validation for the upstream-supported EBS configuration.

Metric	Value
Initial-start self-tests converged	5.000000e+01
Estimated-start self-tests converged	5.000000e+01
Paired fits with objective difference greater than 0.0001	0.00000e+00
Maximum absolute paired objective difference	3.09564e-08
Median across-simulation median SSB percent bias	-1.055%
Median terminal-year SSB percent bias	-0.907%
Fleets with at least one row above 2 percent rejection probability	2.00000e+00
Maximum row-level non-positive probability	6.66814e-02

Table 5: Fleet-level probability of a non-positive index draw under the fitted, untruncated observation distribution.

Fleet	Code	Distribution	Rows	Maximum row probability	Mean row probability	Rows above 2
ATS	4	Lognormal	19	0.00%	0.00%	
ATS_1	6	Lognormal	18	0.00%	0.00%	
AVO	2	Normal	18	3.76%	0.22%	
BTS	3	MVN	42	6.67%	0.17%	
BTS_1	5	Lognormal	42	0.00%	0.00%	
CPUE	7	Normal	12	0.00%	0.00%	

absolute differences of 0.000117% for spawning biomass and 0.000143% for predicted catch. This confirms that the package update preserved the agreement in the population calculations.

The self-test uses the same delivered EBS workbook and supported `MultinomialAFSC` composition criterion as the Rceattle configuration. The observed-data fit reproduced the stored objective of 713.676499, maximum gradient of 0.000044, and `WARN` status from the Hessian condition number of approximately 1.2×10^6. The weak direction remained concentrated in `sel_coff_dev`.

All 50 phased simulations converged with `OK` status when refitted from the original initial values. Repeating the same simulated datasets from the fitted parameter values also produced 50 `OK` fits. Paired objectives agreed within 3.1×10^{-8}, so these simulations showed no start-dependent mode. Median spawning-biomass bias was -1.06% across years and -0.91% in the terminal year. These results validate the revised self-test workflow; they do not rule out the mode previously encountered when fitting the observed data.

One AVO row had a 3.76% non-positive probability and one BTS multivariate-normal row had a 6.67% probability. Those rows exceed the package's 2% warning threshold and should be identified when interpreting recovery. CPUE and the remaining index series had negligible or zero marginal rejection probability.

5.2 OSA Residuals

OSA residuals run on the Rceattle configuration. The aggregate composition fits are generally close, while all three OSA quantile plots are narrower than the standard-normal reference (Figure 4). These diagnostics apply to the fitted `NonParametricPM` selectivity configuration. The failed 2D-AR1 sensitivity and ADMB-to-Rceattle calculation use separate results. The composition diagnostics use `afscOSA` version 0.0.1 and pass Rceattle's internally calculated residuals to `afscOSA::run_osa()`. The residual builder uses complete ages 1–15, with nominal fishery sample sizes and the integer BTS and ATS sample sizes aligned with the modified ADMB model. This follows the current afscOSA interpretation guidance while retaining Rceattle's residual calculation.

Table 6: Mohn's rho for the nine retrospective peels of the Rceattle configuration. Values are the mean relative difference between a peeled estimate in its terminal year and the corresponding estimate from the unpeeled model. Each peel preserves the 2024 data-availability pattern and fixes terminal-year selectivity-at-length parameters to their preceding-year values.

Quantity	Peels	Mohn's rho
Total biomass	9	0.137
Spawning biomass	9	0.274
Recruitment	9	0.139
Fishing mortality	9	-0.157

Following that guidance, the aggregate age-composition fits are the first check for systematic misfit, including patterns that could indicate an unsuitable selectivity form. The red vertical ranges contain 95% of data simulated from the fitted multinomial model conditional on the aggregate sample size and estimated parameter values held constant. They represent conditional simulation variability. The quantile–quantile plots then assess whether the OSA residuals resemble a standard normal distribution. The standard deviation of normalized residuals (SDNR) should be near one, and the displayed tail statistics provide a separate check on extreme residual behavior. Finally, the OSA and Pearson bubble plots are used to look for age, year, or cohort patterns that may suggest the source of misfit. Each statistic and nominal interval contributes to an integrated diagnostic interpretation rather than an automatic model-rejection rule.

The aggregate observed and fitted compositions are generally close, although the survey panels contain some age-specific departures. The OSA distributions are narrower than the standard-normal reference for all three composition sources: SDNR is 0.64 for the fishery, 0.80 for BTS, and 0.52 for ATS. Their tail diagnostics also show shorter-than-expected tails. Together these results indicate underdispersion relative to the fitted multinomial observation model. Model selection requires this information alongside the full suite of diagnostics. The bubble plots should instead be used to identify persistent age, year, or cohort patterns and to guide focused checks of selectivity, composition weighting, and observation-model assumptions. One Pearson residual (8.86) is capped at 6 in the figure, following the package's plotting convention.

5.3 Retrospective patterns

Nine retrospective peels were run from the Rceattle fit. The unpeeled model ends in 2024, and the peeled fits end in 2023 through 2015. Each peel preserves the data-availability pattern of the 2024 model. Thus, fishery age compositions end one year before the peel terminal year, survey series retain their historical sampling schedule and terminal lags, and earlier discontinued series retain their original ending years.

Each peel applies the fitting sequence documented in the appendix. The terminal-year fishery and mirrored catch-per-unit-effort selectivity parameters share the preceding-year parameter values. All nine final fits received `OK` convergence status, with maximum gradients from 3e-05 to 6.7e-05.

The NonParametricPM fishery selectivity is parameterized at length. Consequently, the terminal selectivity-at-length curve exactly equals the preceding-year curve. Rceattle converts that curve to selectivity at age using a year-specific growth and age–length transition matrix, so the reported selectivity-at-age curve can change between those years. This distinction preserves the intended selectivity constraint while retaining each year's biological size-at-age information. These Rceattle fits are separate from the legacy AD Model Builder (ADMB) retrospective files.

The estimated Mohn's rho values are 0.274 for spawning biomass, 0.137 for total biomass, 0.139 for recruitment, and -0.157 for fishing mortality. The positive biomass and recruitment values indicate that the peeled models generally estimate higher quantities at their terminal years than the corresponding unpeeled estimates. The negative fishing-mortality value indicates lower terminal fishing mortality in the peels. Spawning biomass shows the largest average relative difference. The consistent `OK` statuses and small gradients support interpretation of these values as properties of the revised retrospective specification, while the magnitude of

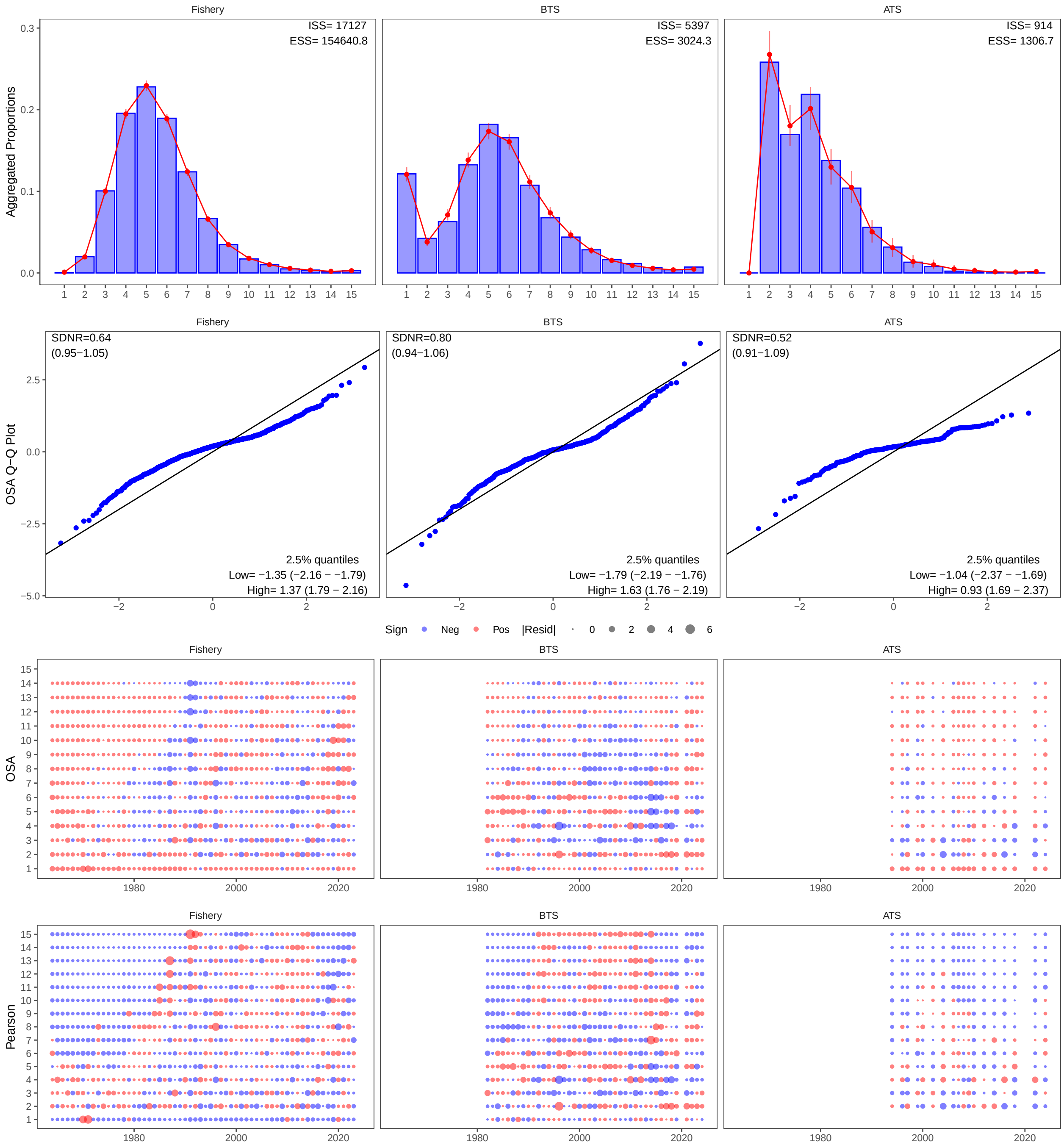

Figure 4: Age-composition diagnostics for the Rceattle fit. The fit uses complete ages 1–15, nominal fishery sample sizes, and BTS and ATS integer sample sizes aligned with the modified ADMB model. Diagnostics were produced with afscOSA for the fishery, bottom trawl survey (BTS), and acoustic-trawl survey (ATS). The top row compares aggregate observed proportions (blue bars) with fitted proportions (red points and lines); red vertical ranges contain 95% of data simulated conditionally from the fitted multinomial model, aggregate sample size, and estimated parameter values held constant. The second row shows quantile–quantile plots of Rceattle's internal one-step-ahead (OSA) residuals with the standard deviation of normalized residuals (SDNR), its expected 95% interval, and lower and upper tail diagnostics. The lower rows show OSA and Pearson residuals by age and year; red and blue indicate positive and negative residuals, and bubble area represents magnitude. Absolute residuals greater than 6 are capped at 6 for display. The fitted model retains ATS 2020; its sample size of one yields zero rounded counts in the afscOSA plotting conversion, so the plotting layer omits that row.

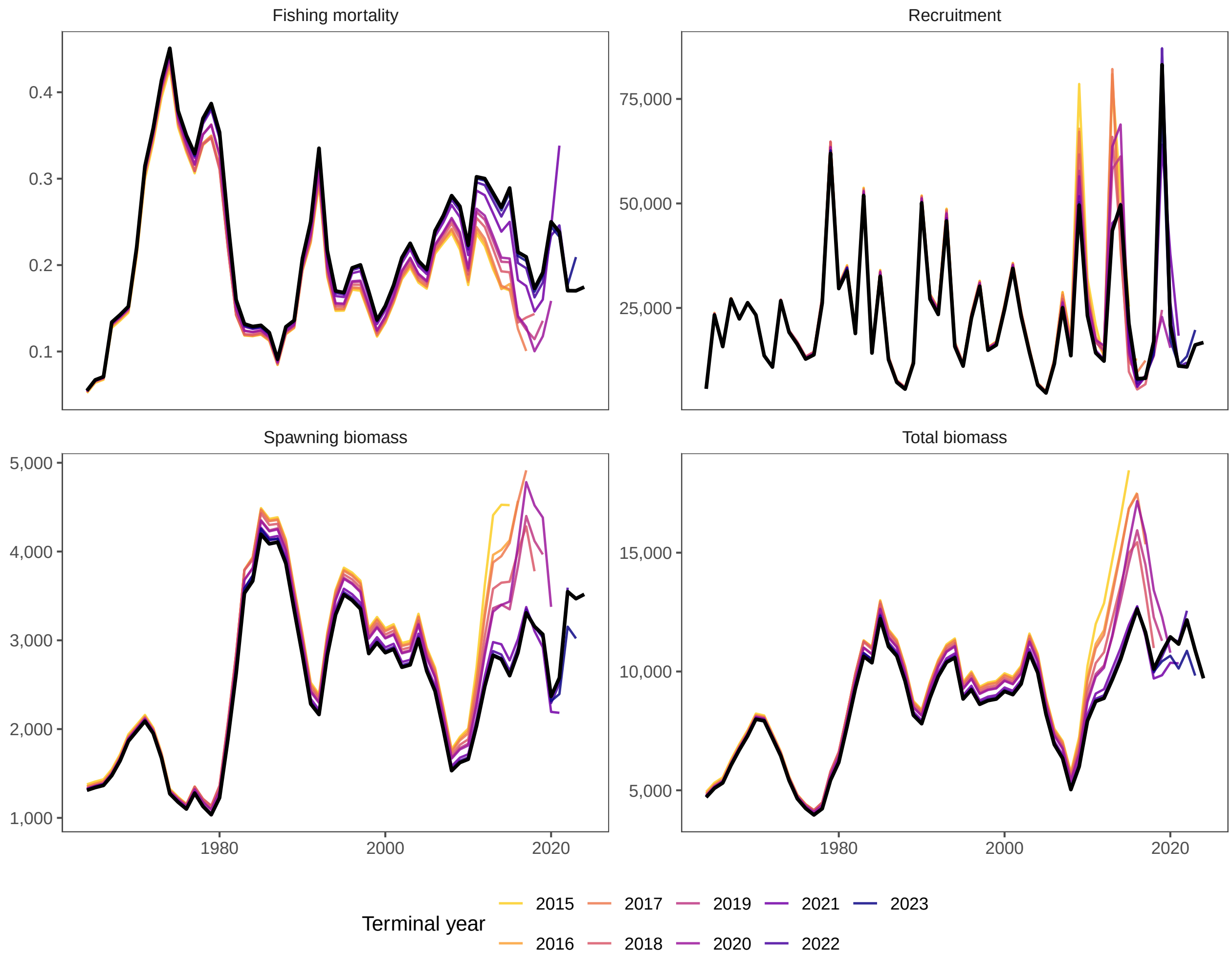

Figure 5: Nine retrospective peels for the Rceattle configuration. The black line is the unpeeled model ending in 2024; colored lines are successive peels ending in 2023 through 2015 and are plotted through each peel's terminal year. Each peel preserves the original data-availability lags and fixes terminal-year selectivity-at-length parameters to their preceding-year values. All nine fits received OK convergence status.

Table 7: Projection files generated from the saved Rceattle fit for the spmR/SPM calculations.

spmR Projection Files

file	role	exists	md5
spm.dat	SPM setup	TRUE	8b4f5e91196e976769547c1c86d1e93d
pm.prj	Rceattle-derived species input	TRUE	301576d1becf07f0172b04f2613dc5da
tacpar.dat	TAC parameters	TRUE	3b02746ed92fe8984e83896029935fba
spm	SPM executable	TRUE	6f52316e14f001e397a765ebfd71f022
age_schedules.csv	Rceattle-derived age schedules	TRUE	90754a0c837abba3a77c4e4481e4df44

spawning-biomass rho motivates continued evaluation of selectivity, survey influence, and recent recruitment estimates.

6 Projections

The saved Rceattle fit supplies every population and fishery input for the Standard Projection Model calculations implemented in `spmR` (Ianelli 2026). The run uses `spmR` version 0.3.0 from pinned main-branch commit `e86fc6aa`. The adapter in `R/write_spmr_projection_inputs.R` writes terminal abundance, natural mortality, maturity, spawning and fishery weights, terminal fishery selectivity, recruitment, and spawning-biomass histories to `pm.prj`. The reproducible runner `R/run_spmr_tier3_projection.R` then applies those inputs to all seven Tier 3 alternatives. The report reads the saved projection products, so every table and figure in this section comes from the same saved Rceattle fit.

6.1 Seven Tier 3 Scenario Results

Catches for 2025 and 2026 are fixed at 1,350 thousand t in every alternative; the alternatives therefore first diverge in 2027 (Table 10). The reported values are simulation means from 1,000 SPM simulations.

6.2 Alternative 2 Fixed-Catch Projection

This additional run applies Alternative 2 and fixes catch at 1,300 thousand t for 2025–2032. The projection horizon includes every fixed-catch year.

6.3 Projection Figures

The figures summarize the 1,000 simulations for each alternative. Lines show simulation medians and ribbons show central 90% intervals.

Table 8: Key settings and source information for the Rceattle-derived SPM projection.

Generated SPM Setup and Source Information

Item	Value
Rceattle source fit	nonparametric_pm
Source checksum	03c339dd48aa7b2627094a3cbab5e43d
Composition likelihood	MultinomialAFSC
Composition sample sizes	fishery nominal; BTS and ATS integer sample sizes aligned with n
spmR version	0.3.0
spmR commit	e86fc6aa
Terminal assessment year	2024
Spawning timing	0.25 year from January 1 (start of April)
Rceattle encoding	spawn_month = 3 elapsed months; 3/12 = 0.25
SPM encoding	Spawnmo = 4 as a one-based calendar-month code; (4 - 1)/12 = 0
Timing compatibility check	Matched: both calculations apply survival through 0.25 of the yea
Spawning-biomass timing convention	Mature spawning biomass after survival through 0.25 of the year:
Projection begin year	2025
Projection years	14
Simulations	1000
Alternatives	1, 2, 3, 4, 5, 6, 7

Table 9: SPM projection alternatives included in the generated setup file.

SPM Projection Alternatives

Alternative	Name	Description
1	Maximum permissible ABC	Maximum permissible Tier 3 ABC harvest rate.
2	Author-specified ABC	Author-specified adjustment to the Tier 3 ABC harvest rate
3	Average recent F	Recent average fishing mortality from the Rceattle-derived i
4	Alternative SPR rate	User-specified SPR rate of 0.60.
5	No fishing	Zero fishing mortality after the fixed-catch years.
6	OFL threshold determination	OFL harvest rate for threshold and status-determination cal
7	Status-determination ramp	Maximum permissible ABC for three years followed by the C

Table 10: Tier 3 projection results for all seven FMP alternatives using inputs generated from the Rceattle EBS pollock model. Catch and biomass quantities are in thousand t; $B/B_{35\%}$ is spawning biomass relative to the Tier 3 $B_{35\%}$ proxy.

Seven Tier 3 Projection Alternatives

		2027					
Alt	Scenario	Catch_2027	ABC_2027	OFL_2027	SSB_2027	F_2027	B_
1	Maximum permissible ABC	1,785	1,785	2,156	2,576	0.330	
2	Author-specified ABC	1,785	1,785	2,156	2,576	0.330	
3	Average recent F	1,163	1,163	2,156	2,662	0.200	
4	Alternative SPR rate	838	838	2,156	2,703	0.140	
5	No fishing	0	0	2,156	2,804	0.000	
6	OFL threshold determination	1,718	1,718	1,718	2,204	0.370	
7	Status-determination ramp	1,621	1,928	1,928	2,379	0.320	

Table 11: Age-specific schedules generated from the terminal-year Rceattle fit and applied in SPM.

Projection Age-Specific Schedules

Age	Spawning_weight	Fishery_weight	Maturity	Natural_mortality	Selectivity
1	0.085	0.025	0.000	0.900	0.000
2	0.196	0.186	0.008	0.450	0.056
3	0.322	0.401	0.289	0.300	0.225
4	0.534	0.667	0.641	0.300	0.521
5	0.621	0.663	0.842	0.300	1.031
6	0.707	0.720	0.901	0.300	1.019
7	0.822	0.791	0.947	0.300	1.001
8	0.946	0.905	0.963	0.300	1.375
9	1.051	1.009	0.970	0.300	1.368
10	1.113	1.061	1.000	0.300	1.360
11	1.178	1.138	1.000	0.300	1.410
12	1.290	1.257	1.000	0.300	1.409
13	1.332	1.267	1.000	0.300	1.409
14	1.367	1.290	1.000	0.300	1.409
15	1.415	1.368	1.000	0.300	1.409

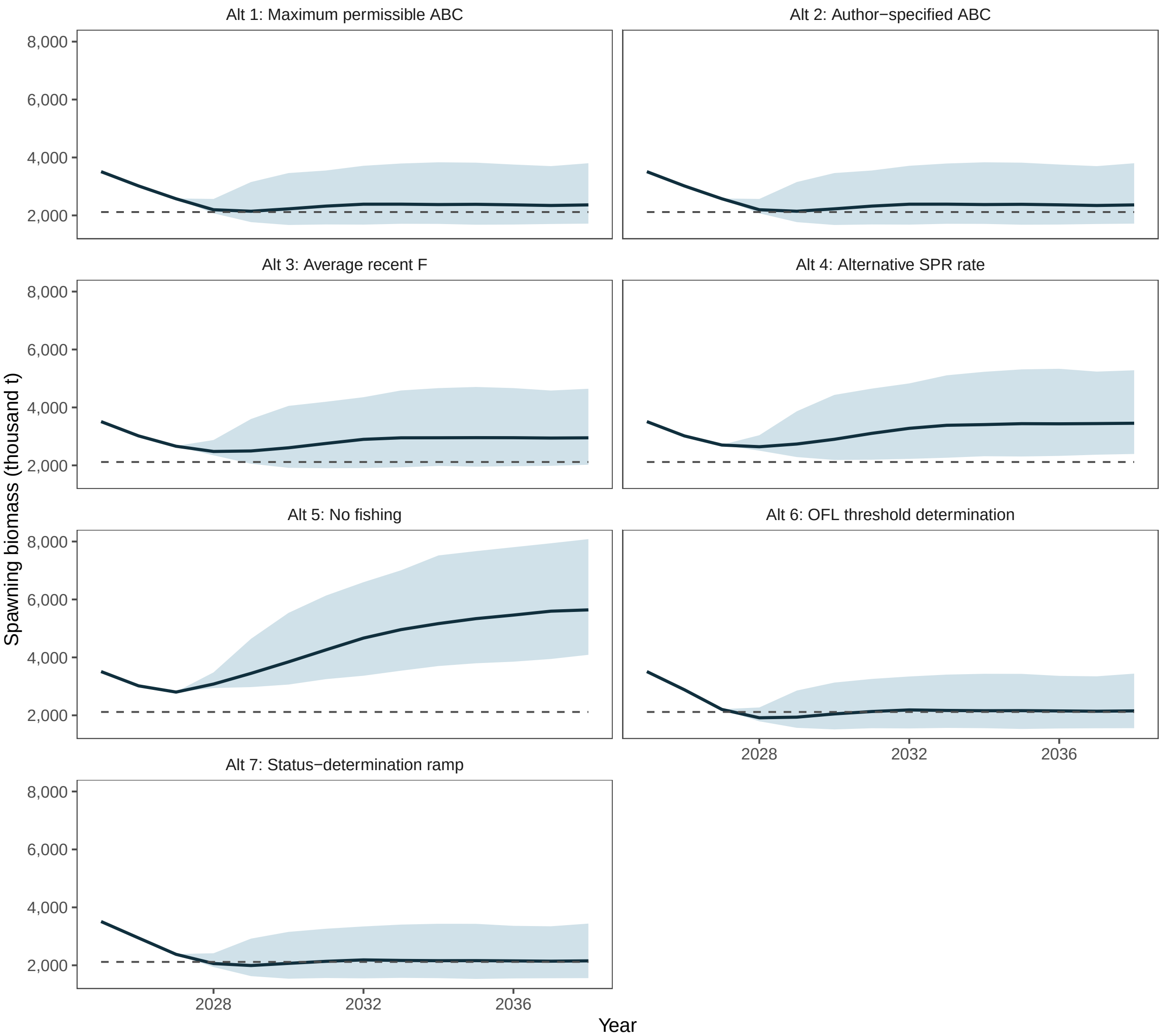

Figure 6: Projected spawning biomass under all seven Tier 3 alternatives. Lines are simulation medians, ribbons span the 5th to 95th percentiles, and dashed lines mark B35%.

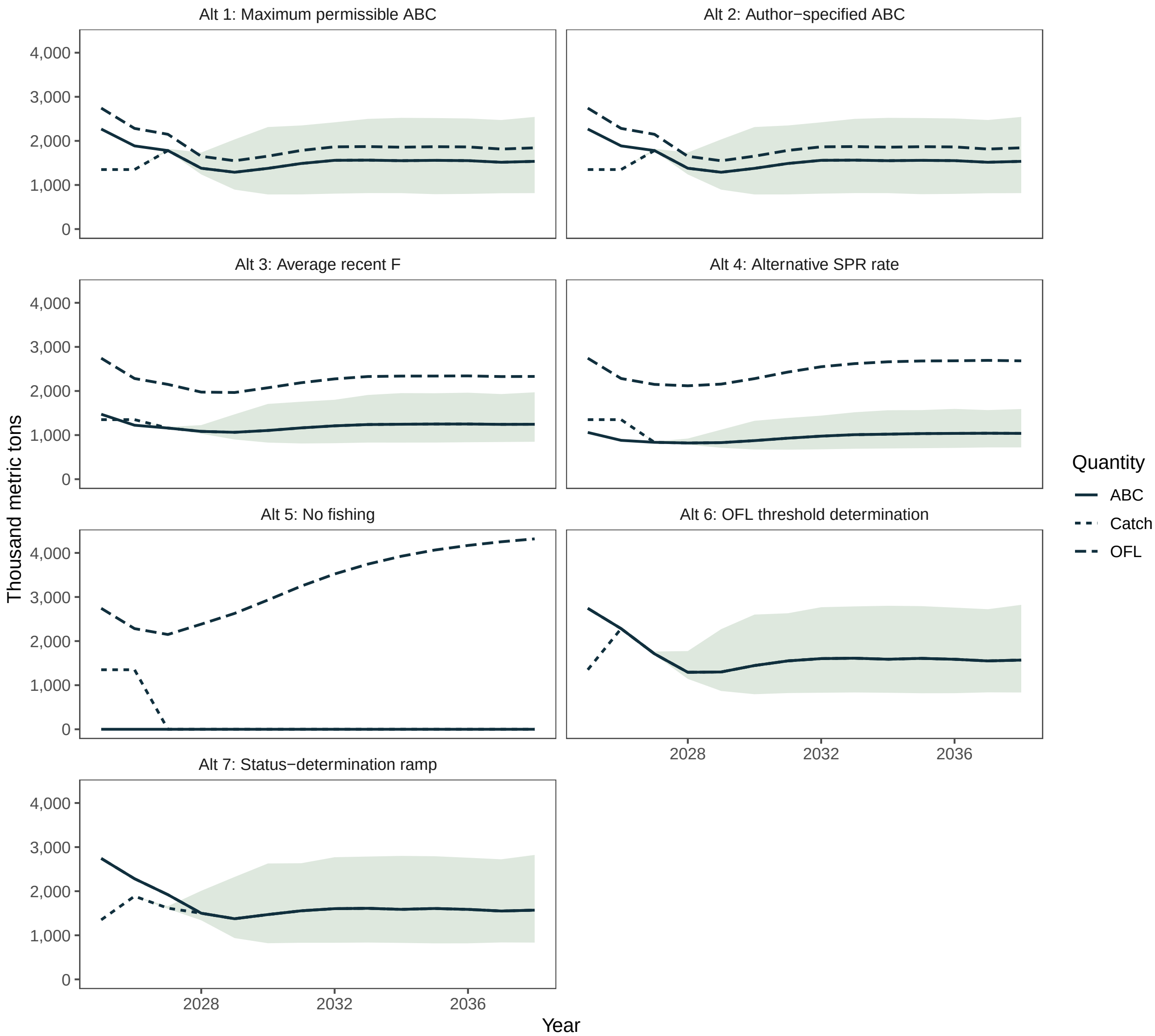

Figure 7: Projected catch, ABC, and OFL under all seven Tier 3 alternatives. Lines are simulation medians and catch ribbons span the 5th to 95th percentiles.

Table 12: Candidate ABC projection given assumed future catches. Catch is fixed at 1,300 thousand t; ABC is the maximum permissible ABC calculated by SPM. $B/B_{35\%}$ is mean spawning biomass relative to the Tier 3 proxy.

Candidate ABC Projection Given Assumed Future Catches

Year	Catch	ABC	OFL	Mean B	$B/B_{35\%}$
2025	1,300	2,268	2,743	3,517	166%
2026	1,300	1,898	2,297	3,044	144%
2027	1,300	1,809	2,184	2,681	127%
2028	1,300	1,646	1,974	2,477	117%
2029	1,300	1,546	1,867	2,472	117%
2030	1,300	1,575	1,906	2,543	120%
2031	1,300	1,621	1,969	2,633	124%
2032	1,300	1,662	2,026	2,720	129%

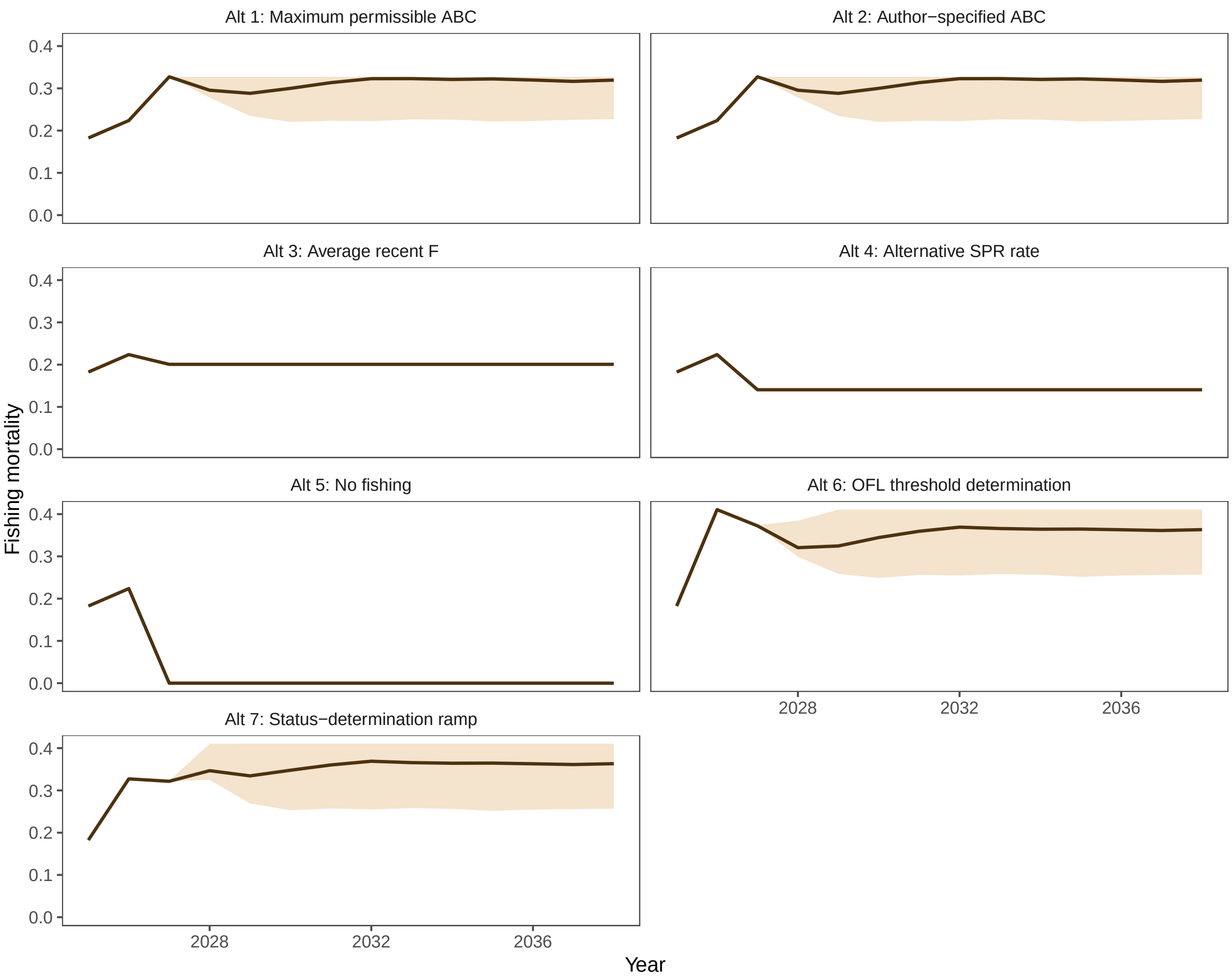

Figure 8: Projected fishing mortality under all seven Tier 3 alternatives. Lines are simulation medians and ribbons span the 5th to 95th percentiles.

Table 13: Mean SPM results by alternative and projection year. Biomass, catch, ABC, and OFL are in thousand t.

SPM Projection Means

Alt	Year	B	Catch	ABC	OFL	F	$B/B_{35\%}$
1	2025	3,511	1,350	2,268	2,743	0.182	166%
1	2026	3,017	1,350	1,886	2,283	0.224	143%
1	2027	2,576	1,785	1,785	2,156	0.327	122%
1	2028	2,244	1,425	1,425	1,709	0.299	106%
1	2029	2,255	1,360	1,360	1,643	0.288	107%
1	2030	2,354	1,443	1,443	1,746	0.291	111%
1	2031	2,433	1,510	1,510	1,828	0.295	115%
1	2032	2,495	1,549	1,549	1,877	0.298	118%
1	2033	2,529	1,580	1,580	1,913	0.299	119%
1	2034	2,524	1,581	1,581	1,914	0.299	119%
1	2035	2,520	1,576	1,576	1,907	0.300	119%
1	2036	2,510	1,571	1,571	1,901	0.299	119%
1	2037	2,505	1,566	1,566	1,896	0.299	118%
1	2038	2,518	1,570	1,570	1,900	0.298	119%
2	2025	3,511	1,350	2,268	2,743	0.182	166%
2	2026	3,017	1,350	1,886	2,283	0.224	143%
2	2027	2,576	1,785	1,785	2,156	0.327	122%
2	2028	2,244	1,425	1,425	1,709	0.299	106%
2	2029	2,255	1,360	1,360	1,643	0.288	107%
2	2030	2,354	1,443	1,443	1,746	0.291	111%
2	2031	2,433	1,510	1,510	1,828	0.295	115%
2	2032	2,495	1,549	1,549	1,877	0.298	118%
2	2033	2,529	1,580	1,580	1,913	0.299	119%
2	2034	2,524	1,581	1,581	1,914	0.299	119%
2	2035	2,520	1,576	1,576	1,907	0.300	119%
2	2036	2,510	1,571	1,571	1,901	0.299	119%
2	2037	2,505	1,566	1,566	1,896	0.299	118%
2	2038	2,518	1,570	1,570	1,900	0.298	119%
3	2025	3,511	1,350	1,472	2,743	0.182	166%
3	2026	3,017	1,350	1,223	2,283	0.224	143%
3	2027	2,662	1,163	1,163	2,156	0.201	126%
3	2028	2,531	1,101	1,101	2,007	0.201	120%
3	2029	2,616	1,107	1,107	1,996	0.201	124%
3	2030	2,757	1,168	1,168	2,113	0.201	130%
3	2031	2,881	1,217	1,217	2,218	0.201	136%
3	2032	2,986	1,251	1,251	2,288	0.201	141%
3	2033	3,059	1,286	1,286	2,359	0.201	145%
3	2034	3,084	1,302	1,302	2,390	0.201	146%
3	2035	3,098	1,305	1,305	2,400	0.201	146%
3	2036	3,099	1,309	1,309	2,407	0.201	146%
3	2037	3,099	1,311	1,311	2,409	0.201	146%
3	2038	3,114	1,313	1,313	2,414	0.201	147%
4	2025	3,511	1,350	1,060	2,743	0.182	166%
4	2026	3,017	1,350	880	2,283	0.224	143%
4	2027	2,703	838	838	2,156	0.141	128%
4	2028	2,694	833	833	2,145	0.141	127%

7 Exploratory DSEM Recruitment Covariates

The DSEM work is an exploratory extension rather than part of the final Rceattle configuration. All assessment comparisons, diagnostics, and projections in this report use the official Rceattle 5.8.1 release. Because that release does not include the DSEM interface, these recruitment runs use a locally labeled 5.8.1.9000 test build: the official 5.8.1 source (7bb69478) with the current DSEM addition (cb7c3f0d). This version record keeps the DSEM experiment separate from the released 5.8.1 package.

Environmentally informed recruitment models benefit from covariates linked to plausible mechanisms, evaluated at relevant temporal and spatial scales, and tested for predictive and management value; otherwise added complexity can produce unstable or spurious relationships (Haltuch et al. 2019). DSEM provides a computational framework for representing simultaneous and lagged ecological pathways in incomplete multivariate time series (Thorson et al. 2024). Causal interpretation also requires checking whether the proposed graph is consistent with its implied conditional-independence relationships (Thorson et al. 2025). A recent Gulf of Alaska pollock application embedded DSEM within a state-space stock assessment and found that an intermediate-complexity causal model reduced unexplained recruitment variance and improved one-year-ahead forecasts, illustrating the value of parsimonious, hypothesis-driven model comparison (Champagnat et al. 2026).

The proof of concept treats recruitment deviations as random effects and aligns covariates with the year in which each cohort enters the model at age 1. The two standardized series are cold-pool extent (42 observations) and a next-year spawning-biomass proxy (47 observations). All four models use the same years, covariate observations, missing-year patterns, and AR1 processes. Only the estimated covariate effects on recruitment change, so AIC can be compared within this set of models.

The conceptual pathway links spawning and environmental conditions to recruitment while identifying the two covariates used here within that broader context (Figure 9). The current proof of concept estimates only the two dashed paths to age-1 recruitment; the solid paths identify mechanisms and covariates for future expansion rather than fitted paths in this analysis.

Table 14: Exploratory DSEM recruitment-covariate models based on Rceattle 5.8.1. Delta AIC is calculated among models using the same years and covariate observations.

Model	AIC	Delta AIC	Maximum gradient	PD Hessian	Status
Next-year SSB proxy	4,091.58	0.00	4.15×10^{-5}	TRUE	WARN
Cold pool + next-year SSB proxy	4,093.22	1.65	5.28×10^{-5}	TRUE	WARN
IID recruitment deviations	4,115.57	23.99	8.60×10^{-5}	TRUE	WARN
Cold pool	4,117.43	25.85	6.27×10^{-5}	TRUE	WARN

The model containing only the next-year spawning-biomass proxy had the lowest AIC (Table 14; Figure 10). Its estimated residual recruitment variance was 24.7% lower than in the IID model; adding cold pool changed that reduction only to 25.1%. The cold-pool recruitment coefficient was small and its interval included zero in both the single-covariate and combined models. The proxy coefficient was negative in both configurations.

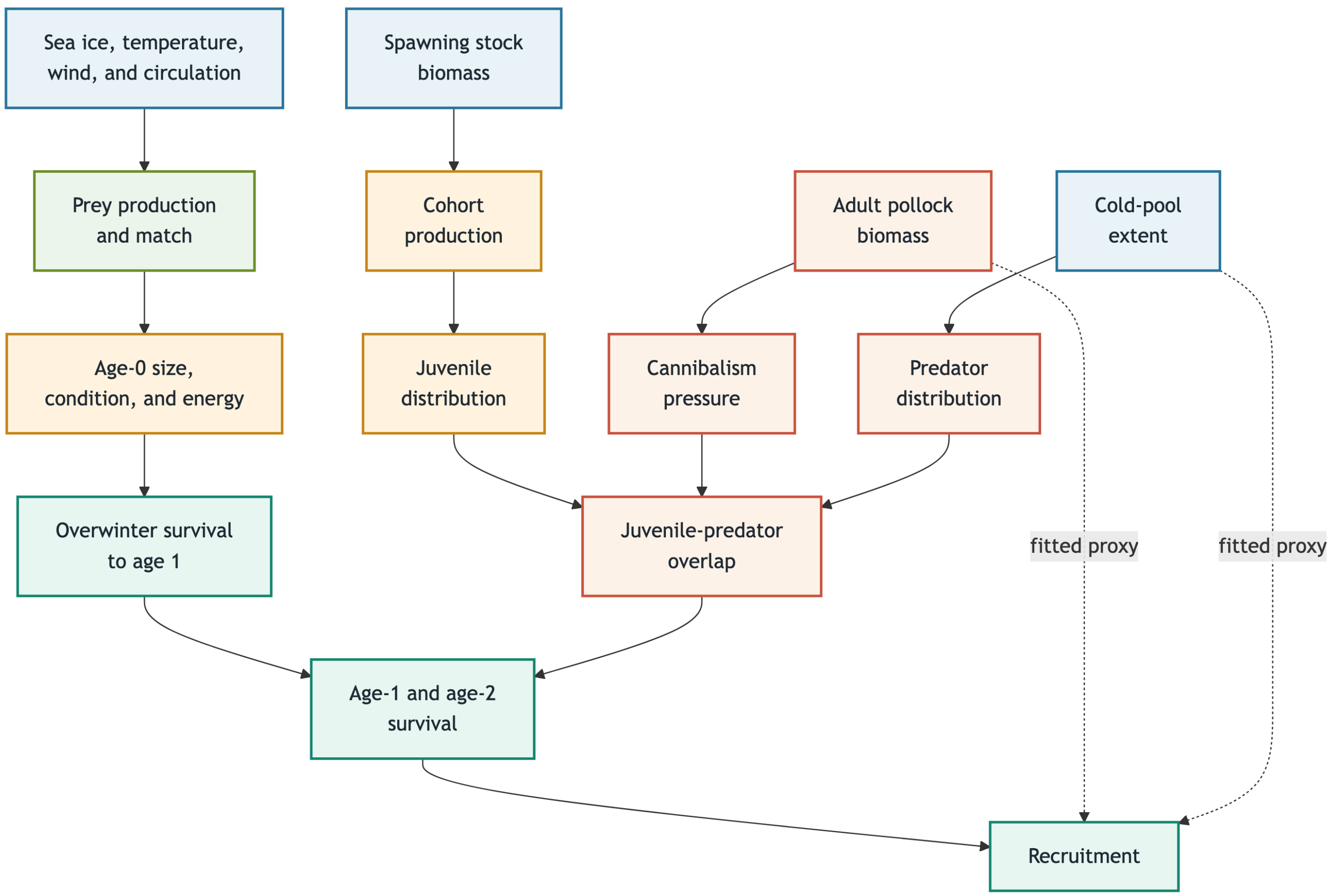

Figure 9: Compact conceptual directed acyclic graph for eastern Bering Sea pollock early survival. Solid arrows summarize broader mechanisms that motivate future covariates. Dashed arrows identify the two direct proxy effects estimated in the current DSEM proof of concept: cold-pool extent and next-year spawning biomass on age-1 recruitment.

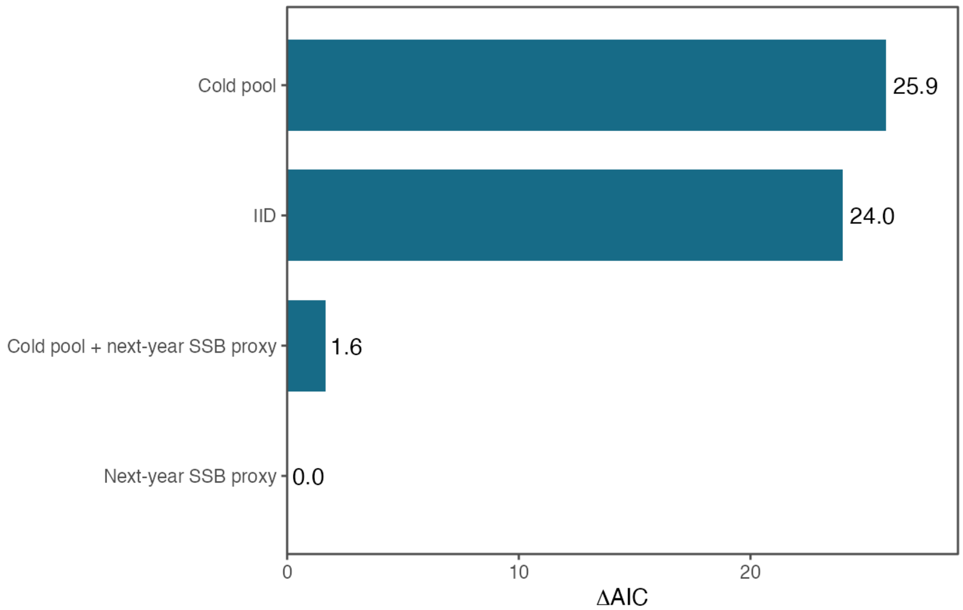

Figure 10: Akaike information criterion differences for the four exploratory DSEM recruitment models. The next-year spawning-biomass proxy model has the lowest AIC; bars show the increase in AIC relative to that model.

Table 15: Standardized DSEM paths to age-1 recruitment. Intervals are estimate plus or minus 1.96 standard errors.

Covariate	Estimate	Standard error	Lower 95%	Upper 95%
Cold pool				
Cold pool	0.040	0.107	-0.169	0.250
Next-year SSB proxy				
Next-year SSB proxy	-0.358	0.072	-0.500	-0.217
Cold pool + next-year SSB proxy				
Cold pool	-0.056	0.095	-0.242	0.130
Next-year SSB proxy	-0.367	0.074	-0.513	-0.221

The next-year spawning-biomass proxy effect is negative and separated from zero, whereas the cold-pool effect remains near zero (Figure 11). The resulting recruitment trajectories remain visually similar despite the AIC differences (Figure 12).

The spawning-biomass proxy belongs to the same pollock population and should not be interpreted as an external causal driver. Its negative association may reflect compensation, shared population dynamics, or model timing. As a randomization check, the analysis replaced both observed series with independent standard-normal values while preserving their missing-year patterns (seed 20260812). None of the randomized recruitment paths improved on the randomized IID model: their delta AIC values ranged from 0.72 to 1.40, and every randomized path interval included zero. This result shows that the same procedure did not identify an effect in these randomized series, but it does not replace out-of-sample validation.

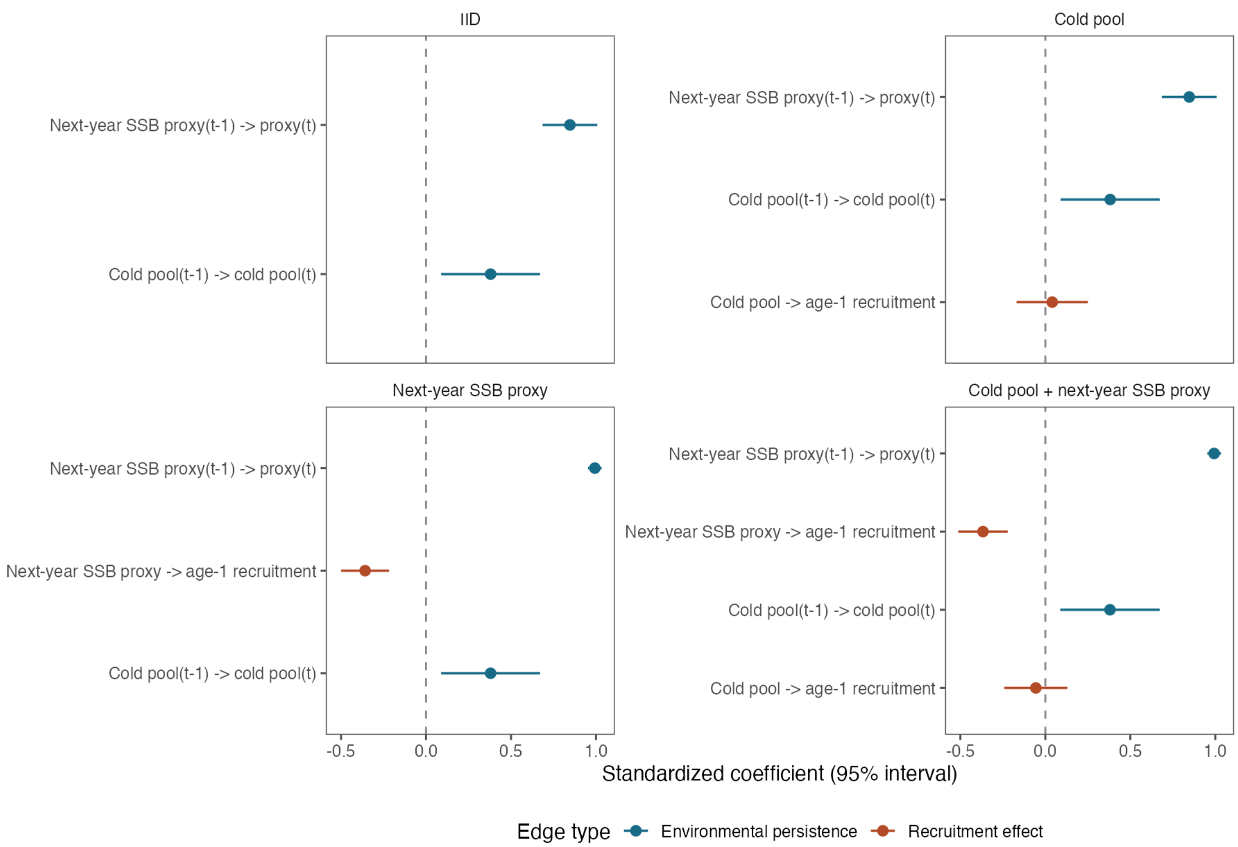

Figure 11: Standardized directed-edge coefficients for the four exploratory DSEM models. Points are estimates and horizontal lines are 95% intervals. Blue edges describe temporal persistence in the observed covariates; orange edges describe direct effects on age-1 recruitment.

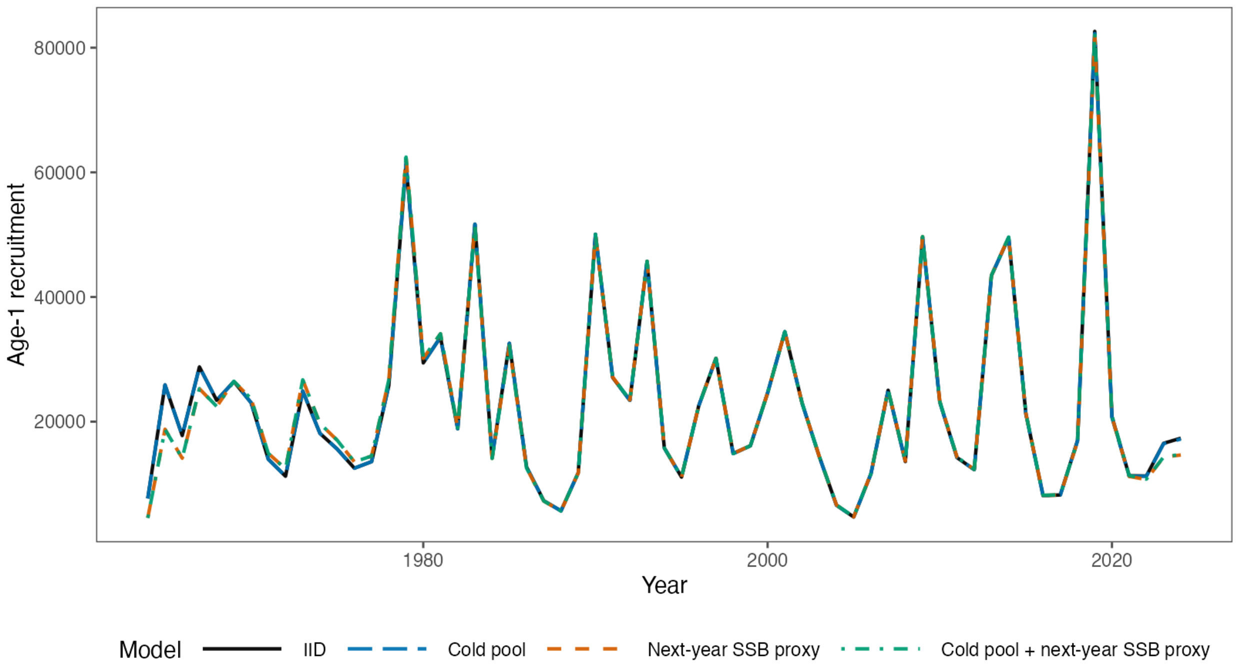

Figure 12: Estimated age-1 recruitment from the IID, cold-pool, next-year spawning-biomass proxy, and combined exploratory DSEM models.

7.1 Adding Covariates

The DSEM script is structured so additional cohort-level covariates can be added without altering the assessment configuration. A useful expansion would:

- state the recruitment mechanism and cohort timing for each candidate before fitting it;
- preserve a common observation-year universe and document original units, standardization, gaps, and data sources;
- screen correlated covariates, then compare IID, single-path, and limited joint-path models rather than searching many combinations;
- repeat missingness-preserving randomized controls and examine sensitivity to observation error, process assumptions, and priors; and
- use blocked or terminal-year holdouts to test prediction before interpreting an in-sample AIC improvement.

This approach can accommodate additional ESP covariates, but the candidate set should remain biologically prespecified and small relative to the available cohort record. DSEM results should remain research products until the paths are stable across timing assumptions, retrospective peels, and predictive checks.

8 Summary

The Rceattle 5.8.1 configuration uses the supported `MultinomialAFSC` composition likelihood, complete ages 1–15, nominal fishery sample sizes, and survey sample sizes aligned with the modified ADMB model. When Rceattle parameters and fishery selectivity are set to the values used in the ADMB model, Rceattle reproduces spawning biomass and predicted catch to numerical precision. This establishes agreement in the tested population-dynamics calculations and parameter mapping. When Rceattle is fit to the data, the trajectories differ more, primarily because the annual selectivity deviations are not estimated stably.

The diagnostic results identify a focused development path for the September 2026 Plan Team evaluation.

The Rceattle fit reaches a small gradient, while Hessian conditioning identifies weak estimability in annual selectivity deviations. Composition OSA residuals remain underdispersed. Nine terminal-year-consistent peels converge successfully and produce positive retrospective bias in spawning biomass, total biomass, and recruitment, together with negative bias in fishing mortality. The historical 2D AR1 sensitivity did not converge and retains its research status. The 50-jitter diagnostic recovered the best solution 48 times and found a poorer mode twice; the age-3+ natural-mortality profile is shallow.

Immediate work should retain the ADMB-to-Rceattle comparison as a repeatable numerical check, evaluate selectivity and survey influence on the retrospective pattern, and refine the model for annual selectivity deviations. The next-assessment workbook should be populated after complete new-year observations arrive. The exploratory DSEM work provides a reproducible approach for adding cohort-aligned covariates, while remaining separate from the final assessment configuration and projection results.

References

Champagnat, Juliette, Cole C. Monnahan, Jane Y. Sullivan, et al. 2026. "Causal Models as a Scientific Framework for Next-Generation Ecosystem and Climate-Linked Stock Assessments." *Fish and Fisheries* 27 (4): 942–59. https://doi.org/10.1111/faf.70092.

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Ianelli, Jim. 2026. *spmR: Standard Projection Model*. http://afsc-assessments.github.io/spmR/.

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Thorson, James T., Cole C. Monnahan, and Lauren A. Rogers. 2025. *Validating Causal Inference in Time Series Models with Conditional-Independence Tests*. EcoEvoRxiv. https://doi.org/10.32942/X29627.

Appendices

A-1 Reproducibility and analysis sequence

The working paper is rendered from saved model and diagnostic products so that building the document does not repeat long model fits. The commands below are run from the project directory, in the order shown when an earlier product is required by a later analysis.

1. The input-preparation script documents the transformations used to prepare the aligned Rceattle workbook; its precursor workbook is regenerated with `Rscript "2024 EBS pollock setup data.R"`. The frozen workbook used for the reported fit is `Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx`. The same script documents the modified ADMB reference rebuild, run as `cd ADMB/m23_rceattle_full && admb pm && ./pm -nox -iprint 150` after the ADMB executable directory has been added to `PATH`.
2. The two-stage fitting script fits the reported Rceattle configuration and the 2D age-by-year AR1 sensitivity with `Rscript "2024 EBS pollock.R"`. The 5.8.1 summary script then generates the fitted-model summaries and standard-multinomial sensitivity using `Rscript R/finalize_rceattle_5_8_1_analysis.R`.

3. The ADMB bridge runner sets the Rceattle parameters and realized selectivity to the ADMB values and saves the trajectory, likelihood, and gradient comparisons with `Rscript R/run_canonical_pm_forward_bridge.R`.
4. The OSA runner generates the saved OSA residuals with `Rscript R/run_canonical_pm_diagnostics.R`. The retrospective script runs the nine terminal-year-consistent peels with `Rscript "2024 EBS pollock two-stage retrospectives.R"`.
5. The 5.8.1 self-test runner generates 50 phased simulations with `Rscript R/run_rceattle_5_8_1_self_test.R`. The restart runner repeats the same simulations from the estimated solution with `Rscript R/run_rceattle_5_8_1_self_test_restart.R`, and the validation summarizer is run with `Rscript R/summarize_rceattle_5_8_1_validation.R`.
6. The projection runner, which uses the Rceattle-to-SPM input writer, generates the seven-scenario and fixed-catch projection products with `Rscript R/run_spmr_tier3_projection.R`.
7. The DSEM script runs the four observed-covariate models with `Rscript "2024 EBS pollock DSEM proof of concept.R"` and the missingness-preserving randomized controls with `Rscript "2024 EBS pollock DSEM proof of concept.R" --randomized`.
8. After these saved products are available, the HTML and PDF are generated together with `quarto render ebswp.qmd`.

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Running under: macOS Tahoe 26.6.1

Matrix products: default
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LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.

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```

Table 16: Files used or referenced by this working paper.

Role	Location	Available	Size
Prepared model-input workbook	Local workbook	TRUE	159.1 KB
BTS covariance input	Local data file	TRUE	12.7 KB
Saved fitted Rceattle objects	Local R serialized-data (RDS) file	TRUE	88036.0 KB
OSA residuals	Local RDS file	TRUE	91.2 KB
OSA diagnostics	Local RDS file	TRUE	1.1 KB
Nine-peel Rceattle retrospective	Local RDS file	TRUE	368664.6 KB
Nine-peel Mohn's rho summary	Local CSV file	TRUE	0.2 KB
Nine-peel retrospective script	Local R script	TRUE	8.1 KB
ADMB-to-Rceattle comparison output	Local RDS file	TRUE	650.8 KB
Rceattle 5.8.1 self-test summary	Local CSV file	TRUE	0.5 KB
Rceattle 5.8.1 truncation diagnostics	Local CSV file	TRUE	0.4 KB
Rceattle 5.8.1 restart comparison	Local CSV file	TRUE	3.8 KB
Rceattle 5.8.1 jitter summary	Local CSV file	TRUE	2.3 KB
Age-3+ natural-mortality profile	Local CSV file	TRUE	0.3 KB
ADMB-to-Rceattle comparison summary	Local comma-separated-values (CSV) file	TRUE	1.0 KB
Rceattle fitting script	Local R script	TRUE	15.7 KB
Diagnostics script	Local R script	TRUE	4.9 KB
Next-assessment update script	Local R script	TRUE	2.6 KB
ADMB comparison script	Local R script	TRUE	10.1 KB
Exploratory DSEM script	Local R script	TRUE	16.8 KB
DSEM model comparison	Local CSV file	TRUE	0.5 KB
DSEM recruitment-path estimates	Local CSV file	TRUE	3.1 KB
Randomized DSEM control	Local CSV file	TRUE	0.5 KB
DSEM AIC comparison figure	Local PNG file	TRUE	24.6 KB
DSEM edge-coefficient figure	Local PNG file	TRUE	101.3 KB
DSEM recruitment-trajectory figure	Local PNG file	TRUE	136.3 KB
Modified ADMB retrospective file	Local ADMB output	TRUE	0.0 KB
Legacy ADMB retrospective file	Local ADMB output	TRUE	0.0 KB

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```

A-2 Model fitting details

The fishery selectivity likelihood has multiple modes, so the reported Rceattle configuration used data-based fishery-selectivity starting values and two fitting stages. The first stage disabled the annual selectivity deviations to establish the selectivity and abundance scale. The second stage restored the deviations and used the first-stage solution as its initial values. This sequence produced a small maximum gradient. The resulting Hessian condition number remained high, and its weakest direction largely involved the annual selectivity deviations (`sel_coff_dev`).

The complete fitting sequence, including the empirical starting values and the two estimation stages, is in the two-stage fitting script and is run with `Rscript "2024 EBS pollock.R"`. The nine-peel implementation is in the retrospective script and is run with `Rscript "2024 EBS pollock two-stage retrospectives.R"`.

The retrospective peels repeat this sequence after recalculating the data-based selectivity starting values for each terminal year. The terminal-year fishery and mirrored catch-per-unit-effort selectivity parameters share the preceding-year parameter values.

A-3 Jitter and natural-mortality profile

The extended-diagnostics runner generates both the 50 jitter fits and the age-3-and-older natural-mortality profile with `Rscript R/run_rceattle_5_8_1_extended_diagnostics.R`.

The 50 phased jitter runs test sensitivity to dispersed starting values. Forty-eight returned to the best objective within numerical precision. Two received `WARN` status and reached an inferior mode with an objective increase of approximately 610.2. This reproduces the pre-existing multimodality and shows why convergence of one run is not sufficient evidence that the global mode was found.

The age-3-and-older natural-mortality profile was evaluated from 0.15 to 0.50. The grid minimum occurred at 0.35, but the objective changed by only 1.27 across the entire grid. The profile is therefore shallow and does not support precise estimation of natural mortality from these data alone.

A-4 Composition-likelihood sensitivity

The 5.8.1 summary script fits the standard-multinomial sensitivity and writes the summaries used here when run with `Rscript R/finalize_rceattle_5_8_1_analysis.R`.

The Rceattle configuration uses the supported `MultinomialAFSC` composition likelihood with a robustness constant of 0.001 added to observed and expected proportions. A standard multinomial fit without this constant was retained as an observation-model sensitivity. Both fits reached small gradients and received `WARN` status because the Hessian's least-identified direction remained in the annual selectivity deviations.

Relative to the `MultinomialAFSC` configuration, the standard multinomial sensitivity changed spawning biomass by 1.60%, recruitment by 0.65%, and total biomass by 1.28% on average in absolute percentage terms.

A-5 2D age-by-year AR1 selectivity sensitivity

The penalized-effects initialization and subsequent Laplace fit are in the two-stage fitting script and are run with `Rscript "2024 EBS pollock.R"`. A separate fixed-scale follow-up can be run with `Rscript "2024 EBS pollock 2D AR1 fixed sd test.R"`; it is a development check and is not the sensitivity summarized below.

The 2D AR1 sensitivity applied an age-by-year field to the shared fishery and catch-per-unit-effort selectivity block over 1964–2024. The age-year deviations were first fit as penalized effects to initialize the field and then

Table 17: Disposition of the Rceattle development components.

Item	Development component	Review result	Interpretation
1	Prepared Rceattle model-input workbook	Used the delivered workbook directly.	Input files are available and reproducible from stored files.
2	Rceattle fit and 2D AR1 sensitivity	The Rceattle fit completed; 2D AR1 received a FAIL convergence status.	The NonParametricPM configuration is the relevant Rceattle comparison; 2D AR1 is diagnostic only.
3	Convergence and model diagnostics	Structured convergence checks, OSA diagnostics, nine retrospective peels, 50 self-tests, 50 jitters, and a natural-mortality profile are saved.	The diagnostics reproduce the selectivity-estimability warning and identify two inferior jitter solutions.
4	Next-assessment data update workflow	Update script review is complete; execution awaits complete new-year observations.	Useful as a template only after complete new-year observations are inserted.
5	ADMB-to-Rceattle comparison diagnostics	Setting Rceattle to the ADMB values reproduces ADMB to rounding-scale precision.	This is the strongest evidence that the ADMB comparison is correct.

Table 18: Rceattle 5.8.1 phased-jitter results for the fitted configuration. Delta objective is measured from the best of 50 runs.

Status	Runs	Minimum delta objective	Maximum delta objective
OK	48	0.000	0.000
WARN	2	610.243	610.243

Table 19: Composition-likelihood sensitivity for the Rceattle 5.8.1 configuration. Raw objectives are shown for run provenance but are not compared because likelihood constants and conventions differ.

Model	Composition likelihood	Objective	Maximum gradient
Rceattle 5.8.1 configuration	MultinomialAFSC	713.676	4.40×10^{-5}
Rceattle 5.8.1 standard multinomial sensitivity	Multinomial	2,840.124	7.57×10^{-5}

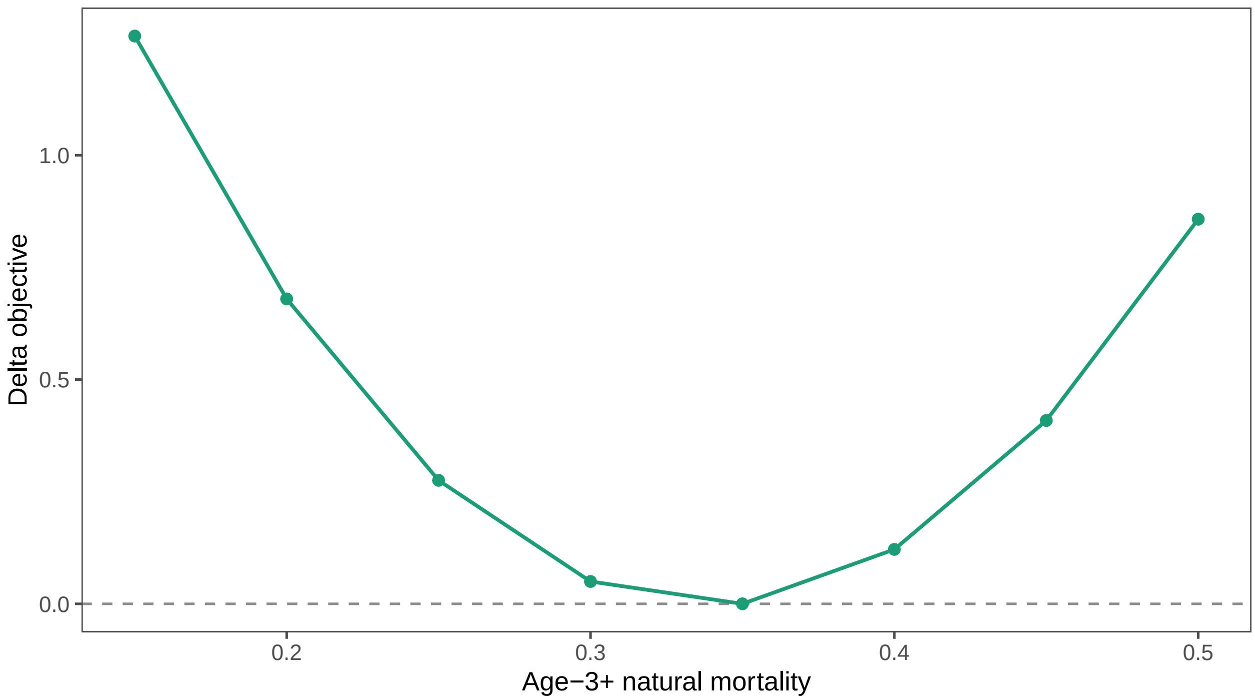

Figure 13: Profile of age-3-and-older natural mortality for the Rceattle 5.8.1 configuration. Points are evaluated grid values and the vertical axis is objective difference from the grid minimum.

Table 20: Diagnostic 2D AR1 fishery-selectivity correlation and scale estimates.

Quantity	Rceattle parameter	Penalized effects	Random-effects fit
Year AR1 correlation	Sel_curve_pen1	0.584	0.912
Age AR1 correlation	Sel_curve_pen2	0.120	0.199
Selectivity deviation scale	sel_dev_log_sd	0.500	0.402

Correlations are on the -1 to 1 scale; the deviation scale is a positive standard deviation. The fit failed convergence checks.

integrated as random effects with a Laplace approximation. The initialized Laplace fit failed the gradient, Hessian, and bound checks.

The failed random-effects fit had a maximum marginal gradient of 19.7 on `rec_pars`, a non-invertible Hessian, 61 `log_F` parameters at their configured bounds, and `FAIL` convergence status. These estimates remain diagnostic.

A-6 Acronyms

This appendix follows the glossary convention in the Alaska Stock Assessment Report (`asar`) package (Schiano et al. 2026). The project glossary uses ASAR's `\newacronym` format and adds terms specific to this Rceattle comparison. The quoted label "PM" is a model name.

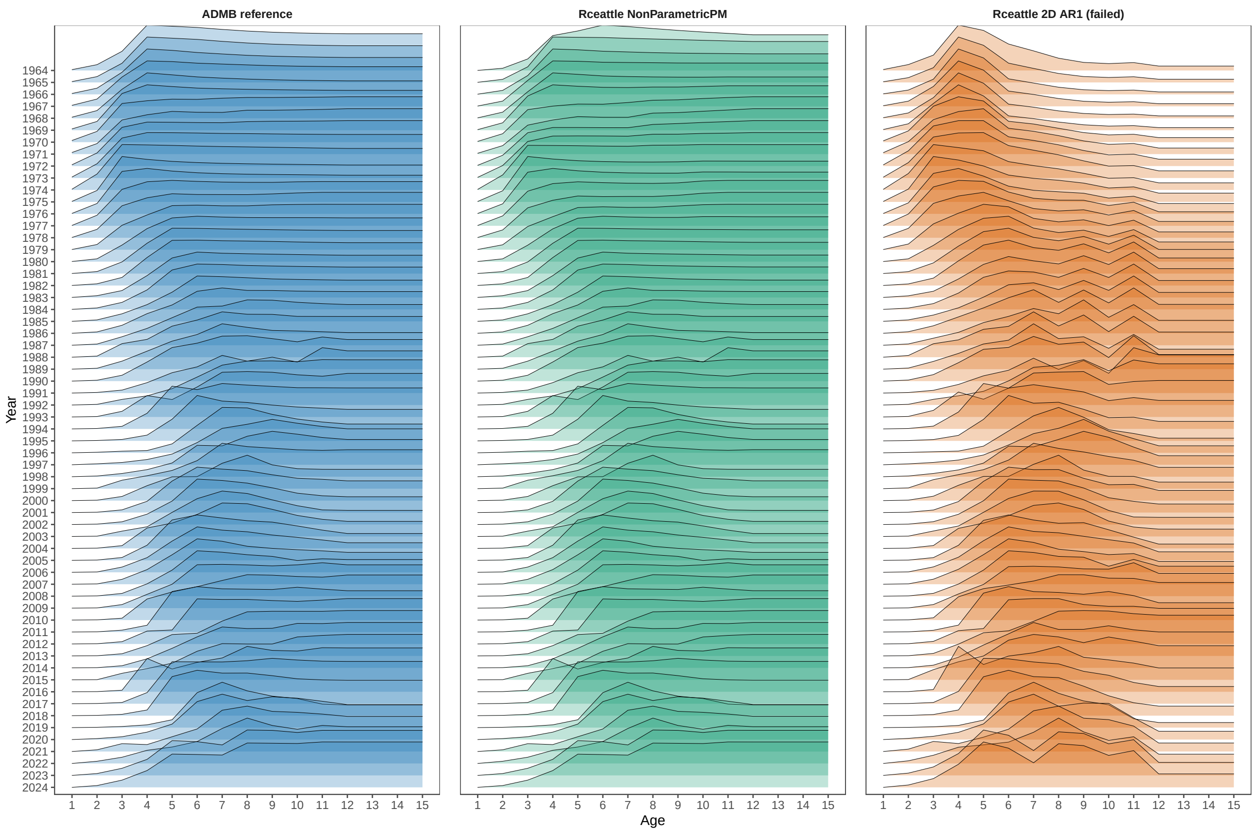

Figure 14: Development comparison of relative fishery selectivity for ADMB, the Rceattle configuration, and the failed 2D age-by-year AR1 sensitivity.

Table 21: Acronyms used in this working paper.

Acronym	Definition
ADMB	AD Model Builder
AIC	Akaike information criterion
AR1	first-order autoregressive
ATS	acoustic-trawl survey
AVO	acoustic vessels of opportunity
BTS	bottom trawl survey
CIE	Center for Independent Experts
CPUE	catch per unit effort
CSV	comma-separated values
DAG	directed acyclic graph
DSEM	dynamic structural equation modeling
ESP	ecosystem and socioeconomic profile
HTML	HyperText Markup Language
IID	independent and identically distributed
MCMC	Markov chain Monte Carlo
MVN	multivariate normal
OSA	one-step-ahead
RDS	R serialized data
RTMB	R Template Model Builder
SAFE	Stock Assessment and Fishery Evaluation
SDNR	standard deviation of normalized residuals
SEM	structural equation model
SPM	Standard Projection Model
SST	sea surface temperature